

USED CAR PRICE ANALYSIS

-A.S.S.V.Akshai

TABLE OF CONTENTS:

1. Introduction
2. Problem Statement
3. Objectives
4. Methodology
 - Data Collection
 - Data Cleaning
 - Data Normalization
 - Data Visualization
5. Results
6. Key Findings
7. Challenges Faced
8. Conclusion
9. References

INTRODUCTION

The used car market in India has grown rapidly, offering an affordable alternative for consumers looking for high-quality vehicles at lower prices. Platforms like Cars24 provide a comprehensive listing of used cars, including various car companies, fuel types, and transmission systems. This project analyzes the variation in car prices across different factors to help consumers make better decisions when purchasing used cars.

However, the price of used cars can vary significantly based on numerous factors such as:

- Company Name
- Fuel Type
- Distance Driven
- Prices

PROBLEM STATEMENT

Understanding the factors that influence the price of used cars is critical for both buyers and sellers. This project seeks to answer the following:

- How do car prices vary based on company, fuel type and Distance Driven?
- What are the best value cars in terms of price and quality?
- What is the impact of car age and distance driven on the price of used cars?

OBJECTIVES

The primary objectives of this project are:

- To analyse car price variation across different companies in the used car market.
- To identify cars that provide good quality at a lower price (value-for-money cars).
- To study the impact of factors such as car age, distance driven, fuel type and transmission on car prices.
- To study the company-wise car availability in each city.

METHODOLOGY

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and visualization. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

1. Data Collection:

The first step of the methodology involves gathering data on used cars. For this project, data is collected using web scraping from the online platform Cars24, a popular marketplace for buying and selling used cars.

- **Tools Used:** Python libraries such as Selenium, BeautifulSoup, and Pandas are utilized for web scraping and data handling.
- **Process:**
 - Selenium is used to automate the process of visiting the Cars24 website and navigating through the car listings.
 - Once the data is loaded, BeautifulSoup parses the HTML content and extracts relevant data fields such as Car name and model, Car Company, Fuel type (Petrol, Diesel, CNG), Distance Driven (Distance covered), Price (in lakhs).
 - The extracted data is then stored in structured formats (like Pandas Data Frame) for further analysis.

2. Data Cleaning:

After collecting the raw data, the next step is Data cleaning. The goal of this phase is to ensure the dataset is free from errors, missing values, and irrelevant information.

Cleaning Steps:

- Removing the Duplicates and the Null Values.
- Replacing the missing values with relevant placeholders.
- Converting the Datatypes of the Columns.
- Run the basic EDA (Exploratory Data Analysis) using python.

3. Data Normalization:

Normalizing prices and other numeric fields to standard formats. For example, prices are converted to a uniform format (in lakhs), and also Distance Driven values are converted into the numerical fields.

4. Data Visualization:

After Cleaning the dataset, the next step is Data visualization, which helps in understanding the relationships between various factors like car company, price, Distance Driven and Fuel type.

Tools Used:

➤ Power BI:

For creating interactive and dynamic dashboards that allow users to filter data and explore trends across different car companies, fuel types, and etc.

➤ Creating Dashboard:

Dashboard 1: Used Car Market Analysis

Purpose:

To understand the overall used car market trend

Key Visuals:

- Total cars available
- Average price, EMI, car age, and distance driven
- Top brand by inventory
- Price depreciation trend by car age
- City, fuel type, and company slicers

Insights:

- Car prices decrease steadily as vehicle age increases
- Petrol cars dominate the used car market
- Certain brands consistently lead market inventory

Dashboard 2: Brand & Pricing Analysis

Purpose:

To compare used car brands based on pricing, affordability, and usage.

Key Visuals:

- Certain brands consistently lead market inventory
- Average price by brand
- Average EMI by brand
- Brand-wise value comparison matrix
- Car Age Group slicers

Insights:

- Budget brands offer lower prices and affordable EMI options
- Premium brands have significantly higher prices and EMIs
- Some brands provide better value despite higher usage

Dashboard 3: Customer Decision Dashboard

Purpose:

To help customers identify the best value used cars based on their preferences.

Key Visuals

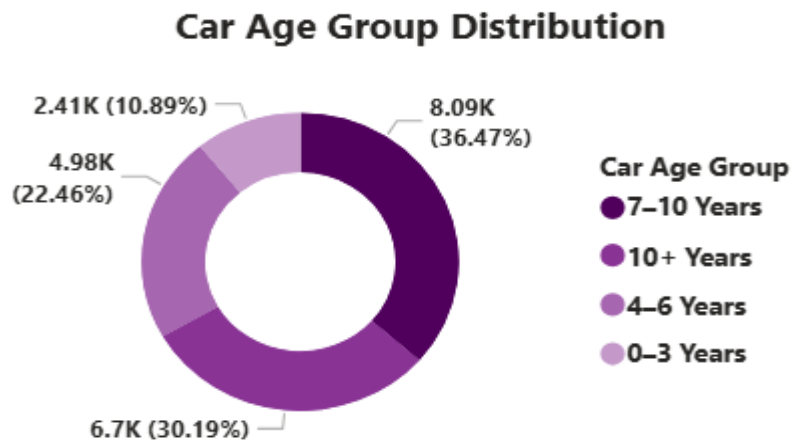
- Car age group distribution
- Price segment distribution
- City, budget, fuel type, and transmission slicers

Insights:

- Buyers can easily filter cars based on budget and preferences
- Low usage cars offer better long-term value
- Most available cars fall within mid-range price segments

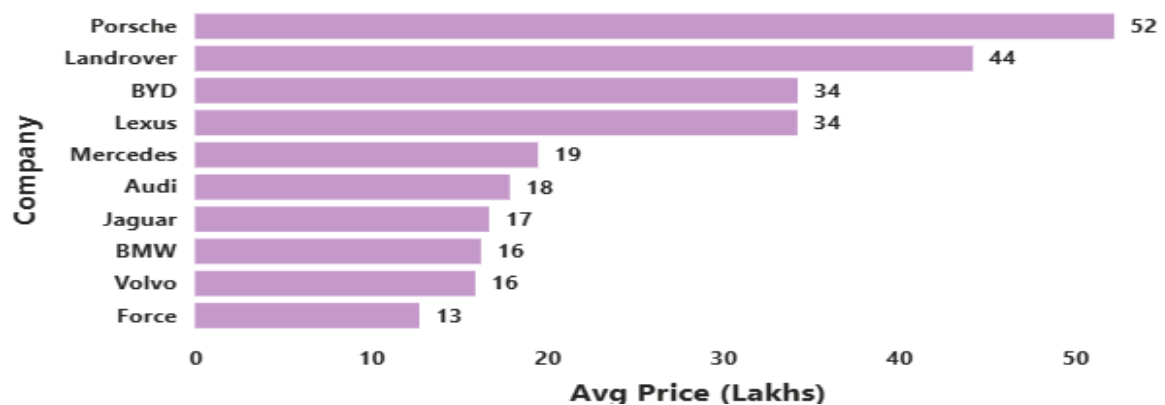
RESULTS

➤ Car Age Group Distribution:



➤ Average Price by Company

Average Price by Company (Top 10)

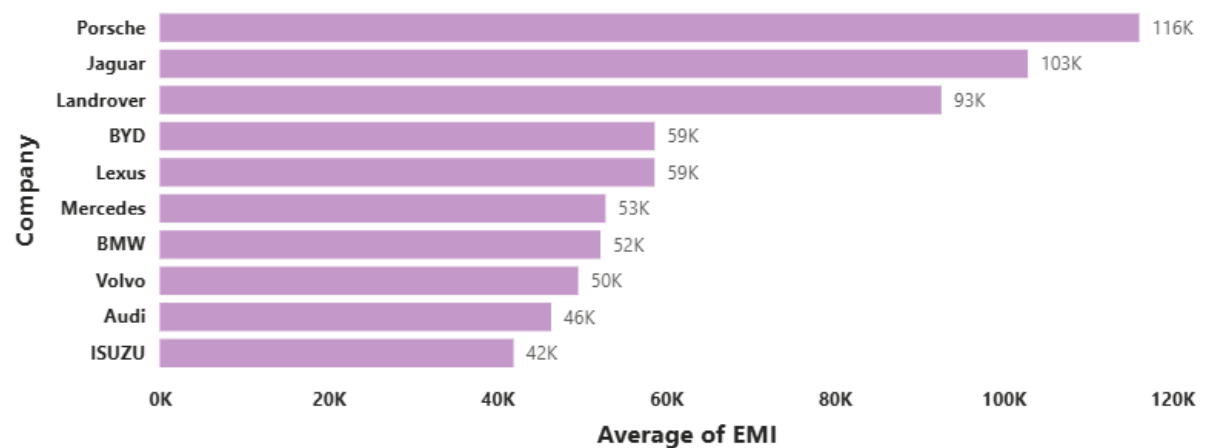


➤ Brand-Wise Value Comparison:

Brand-wise Value Comparison			
Company	Avg Price (Lakhs)	Average of Distance Driven	Average of EMI
Chevrolet	1.45	78807.19	10483.81
Fiat	1.58	100515.37	9795.80
Datsun	1.80	56513.33	3694.44
Renault	3.46	63386.03	8377.02
Ford	3.88	89794.27	13423.95
Maruti	3.92	78011.83	10715.18
Nissan	3.93	66767.96	9766.74
Honda	3.93	80093.85	11168.58
Mitsubishi	4.08	112047.33	25666.00
Hyundai	4.56	73054.46	12081.88
Volkswagen	4.68	82707.37	12662.54
Ssangyong	5.31	106303.33	22334.83
Tata	6.59	60198.36	13654.38
Toyota	7.67	96642.98	23227.46
Skoda	7.78	78846.26	18285.14
Mahindra	9.01	72820.45	22441.85
KIA	9.89	56294.34	18217.44
ISUZU	10.36	137136.64	41947.07
Jeep	11.44	77007.97	24814.27
CITROEN	12.04	43791.27	21666.19
MG	12.09	54503.88	22219.38
Force	12.76	22059.83	23063.00
Volvo	15.94	133993.30	49604.73
BMW	16.26	89149.59	52236.56
Jaguar	16.72	63124.86	102899.86
Audi	17.90	82186.96	46393.73
Mercedes	19.50	71983.82	52838.16
Lexus	34.25	43930.00	58644.00

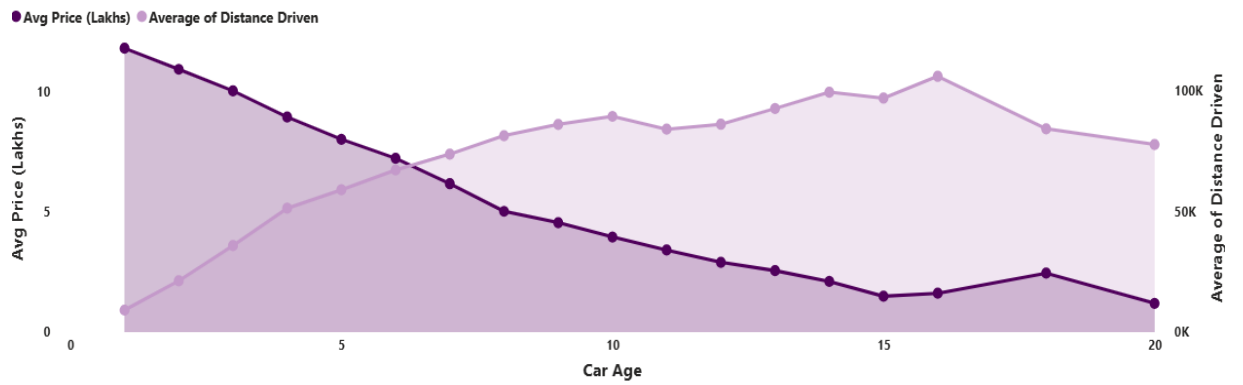
➤ Average of EMI by Company:

Average of EMI by Company



➤ Average Price & Average Distance Driven by Car Age:

Avg Price & Average of Distance Driven by Car Age



KEY FINDINGS

➤ **Car Price Variation by Company:**

Porsche has the highest average car price (around ₹52 Lakhs), mass market companies such as Maruti, Tata, and Hyundai fall into the lower average price range, making them more affordable.

➤ **No. of Used Cars by Company:**

Maruti dominates the used car market other companies like Honda, Mahindra, Renault, Ford, and KIA have comparatively lower representation.

➤ **Average Price vs Car Age:**

Average car price decreases consistently as car age increases, confirming depreciation over time. The steepest price drop is observed between 2 to 8 years of car age.

➤ **Brand Wise Value Comparison:**

Chevrolet, Fiat, and Datsun offer the lowest average prices (₹1.4–1.8 lakhs). Mid-range brands like Maruti, Hyundai, Honda, Nissan, and Renault fall in the ₹3–5 lakh price range balance price and reliability. Premium brands have high cost despite similar usage.

CHALLENGES FACED

- Initially, Data extraction using BeautifulSoup was unsuccessful due to the dynamic nature of the website and JavaScript-rendered content. To overcome this limitation, Selenium was used to automate browser interactions and successfully fetch the required data.
- The dataset contained missing values, inconsistent data formats (such as price, distance driven, and EMI), and incorrect or unrealistic entries. These issues were overcome by performing data cleaning techniques including handling missing values, standardizing units and formats, correcting inconsistencies, and removing or treating invalid outliers to ensure data accuracy and reliability.
- **Incomplete Data Loading Despite Auto-Scrolling:**

Even after implementing Selenium-based auto-scrolling, the webpage initially appeared to load all available listings. However, upon verification, only a limited number of records (approximately 20–30 cars) were actually captured. This issue occurred due to lazy loading and dynamic rendering mechanisms used by the website, requiring additional validation logic and repeated scrolling checks to ensure complete data extraction.

```
df.head(5)
```

	Car Name	Company	Year of Manufacture	Distance Driven	Fuel Type	Gear Type	EMI	Price	Car Age	City
0	Tata TIGOR	Tata	2021	36206	Petrol	Manual	10060	570000.0	5	Bangalore
1	Maruti Wagon R 1.0	Maruti	2017	43965	Petrol	Auto	7484	383000.0	9	Bangalore
2	Renault Kwid	Renault	2017	45145	Petrol	Manual	5645	289000.0	9	Bangalore
3	Maruti Wagon R 1.0	Maruti	2017	49218	Petrol	Auto	7147	366000.0	9	Bangalore
4	Volkswagen Ameo	Volkswagen	2016	67291	Petrol	Manual	9518	428000.0	10	Bangalore

```
cars = soup.find_all('div',class_="styles_outer__NTVth")
len(cars)
```

CONCLUSION

This analysis shows that car price is mainly influenced by company, vehicle age, and usage, with clear depreciation as car age and distance driven increase. Petrol cars dominate the market and most used cars fall in the mid-price and mid-age segments. By understanding factors such as company, car age, usage, and customer demand, sellers can price vehicles more accurately. These insights help improve sales planning, meet customer expectations, and stay competitive in the used car market.

REFERENCES

- **Data Sources:** Cars24 Website
- **Tools Used:** Python (Selenium, BeautifulSoup, Jupyter Notebook), Power BI.